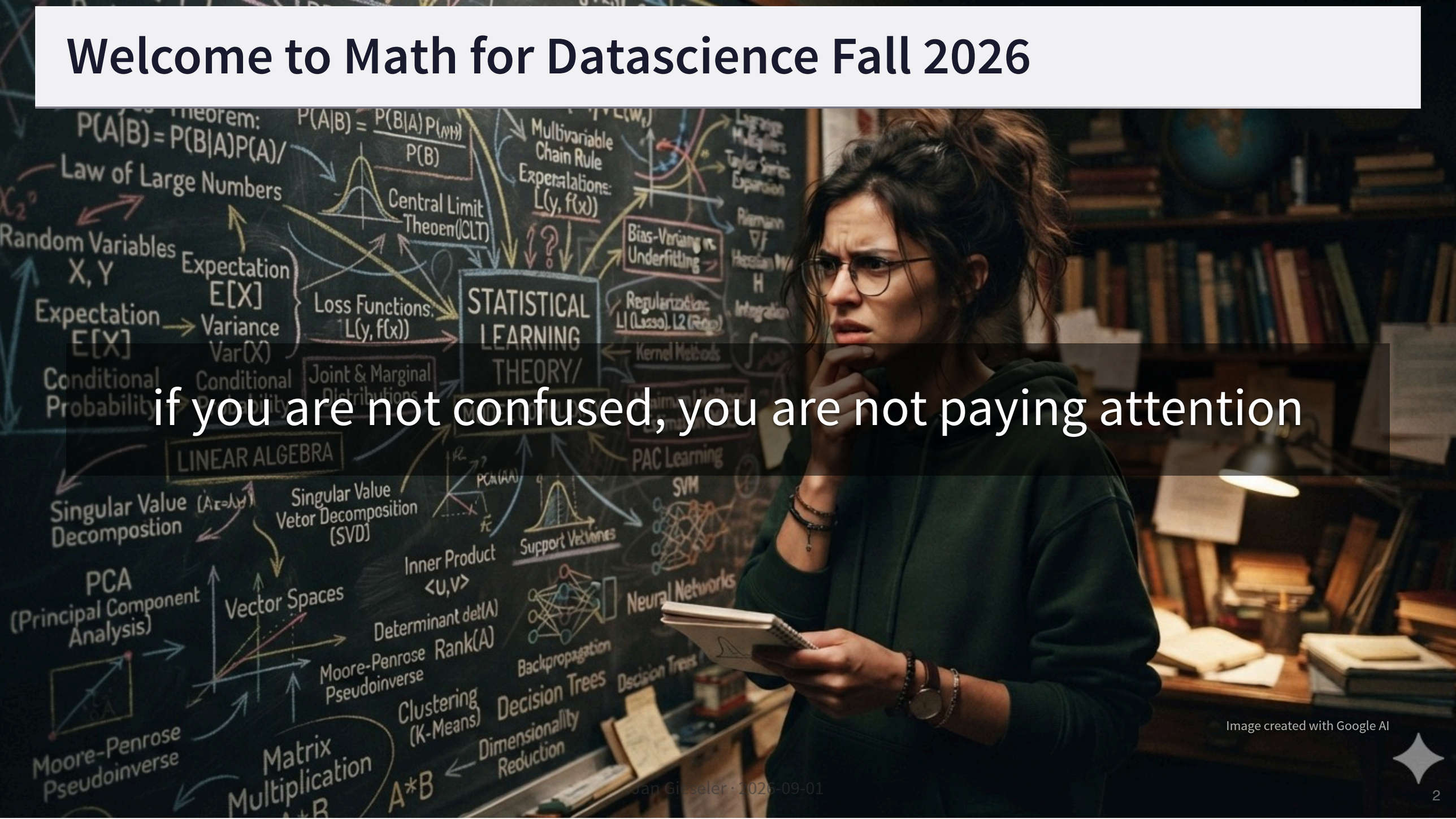
Lecture 1: Probability Theory

Math for Data Science · 2026

Jan Gieseler

2026-09-01

Welcome to Math for Datascience Fall 2026



if you are not confused, you are not paying attention

Image created with Google AI

Math for Datascience Fall 2026

🌐 https://hertie-maths-data-science-f2026.github.io

- GitHub (lecture materials, code, assignments, etc.)
- Moodle (everything else)

Me

🙋‍♂️ I am Jan Gieseler /ˈjan ˈgiːzələr/

✉️ j.gieseler@hertie-school.org

💼 physicist | data scientist / team manager / lead dev @ IAV GmbH



Jan

An Iterative Workflow for Success

1. Do the reading before Tuesday lecture
    - But if it’s very challenging, don’t struggle. Skimming is ok.
2. Come to Tuesday lecture
    - Listen to lecture & ask questions
    - Attendance is mandatory
3. Before your lab
    - Review the slides and do the reading again, this time carefully focusing on the parts covered in slides
    - I will never hold you accountable for material not mentioned or covered in slides.
4. Come to lab

The labs

- Attendance is also mandatory
- Opportunity to ask more in-depth questions; we apply and extend concepts from lecture, often with code
- Use this as an opportunity for self practice.



Camilo

self practice

This is where the learning happens!

Attendance Policy

- That said, things inevitably happen - people travel, get sick, etc.
- You can miss up to two sessions, no questions asked.
- Please do not contact me to inform me of your absence for the first 2 sessions.
- If you have to miss more (3-4 sessions): you need an excused absence (e.g. doctor’s note) submitted to Jan Gieseler (j.gieseler@hertie-school.org).
- If you miss a fifth session, you automatically fail the class
- This policy applies in parallel to both lecture and labs
- Switching labs: ok a few times (email TAs in advance); for permanent switch, contact CA

Assignments

Three take-home group problem sets and one individual (3 x 10% + 15%, 45%)

- Deadlines evenly spaced throughout the semester
- Work in assigned groups of 3 and type up one solution set per group
- Designed to take time and thought; please start early
- Please type solutions neatly and show all work
- A good opportunity to learn LATEX or similar to write nicely formated documents

Assignments

5 short in-class quizzes (5% each, 25%)

- Taken at the beginning of lecture in sessions 3, 5, 7, 9, and 11
- Covering material up to and including the prior week's lecture (e.g., quiz in session 3 covers sessions 1-2, quiz in session 5 covers sessions 3-4…)
- Pen and paper, no coding

Assignments

Final exam (30%)

- Pen and paper, no coding

Hertie School Grading Guidelines

In order to avoid grade inflation and grading inconsistency, we ask instructors to apply the following interpretation of our grading scale:

- Course grades usually range from 70% to 95%
- Aim at a median of 85% which is a good grade
- Only grade beyond 90% if your expectations are greatly exceeded
- Only grade below 70% if you are very disappointed
- Grades above 95% and below 70% should be exceptional (top / bottom 5% of students)

If we do not achieve this distribution naturally, we will curve the grades to get there at the end of the semester.

Coding

- Although this is a conceptual class, supplementing math with code:
  1. Firms up understanding
  2. Connects concepts to real-world problems
- But I am aware this is yet another skill to master, and will provide as much guidance as possible
- You will not be assessed on the quality of your code here - only your understanding of the ideas
- But it’s always good to practice! With AI coding has never been this easy.

Policies on AI Assistance

- This is in part a coding class, and AIs such as ChatGPT / Claude / Mistral / Gemini are a powerful coding assistants (try it if you haven't!). So it would be silly for us to ban its use.
- But, you need to use AI tools responsibly.
- Note that AI policies are specific to this class and other classes may differ. Always follow the professor's guidance.

Be always be wary of mistakes!

The main skill you learn is mathematical and critical thinking. If you leave the thinking to the AI, you will not learn and you will not be able to spot mistakes.

AI in This Course

Use it for

- Search engine
- Boilerplate code
- Explain code / concepts - but always be wary of mistakes!
- Worked examples
- Decoding error messages
- Checking work you already did
- Translating

Do not use it for

- Producing what you hand in - a solution you cannot re-derive is not yours
- The first pass at a derivation - the struggle *is* the learning
- Anything you cannot verify - it invents proofs, identities and citations that look right
- Exams - pen and paper!
- Numbers you then trust - check them yourself

Let’s put our challenges on the table

1. This is an ambitious class.

- Introductory probability, linear algebra, calculus, and statistics are each one semester of university-level math
- Thus the ideal audience for this class has already had some exposure to these topics, and this is review

2. In practice, there will be a wide range of prior exposure to these concepts in this room

- There is a path to success for everyone
- Students with low exposure: there will be clearly articulated and achievable minimal standards of success
- Students with high exposure: material is presented with a new data science/social science perspective, and there will be opportunities for more challenging work
- What I need from you: communication, patience, hard work, and mutual support

Class etiquette

- Learning can be challenging and might lead you out of your comfort zone.
- If you have problems with the pace of the course, let me and the TAs know.
- I expect your commitment to the class, but I do not want anyone to fail.
- You are all genuinely interested in data science. But there is also considerable variation in your backgrounds.
- This is how we like it! Some sessions will be more informative for you than others.
- If you feel bored, look out for and help others, or explore other corners of data science you don’t know yet.
- Be respectful to each other, all the time. This includes the TAs and me.
- Ask questions whenever you feel the need to do so!

Getting Help

For issues relating to	Please contact
Attendance	Your TA
Accommodations	Your TA
Questions about the material	Your TA
Grading questions	Your TA
Scheduling appointments with me during office hours	Lecturer*
General concerns about the course	Lecturer*
Lecture clarification or follow-up	Lecturer*
Research, further study	Lecturer*

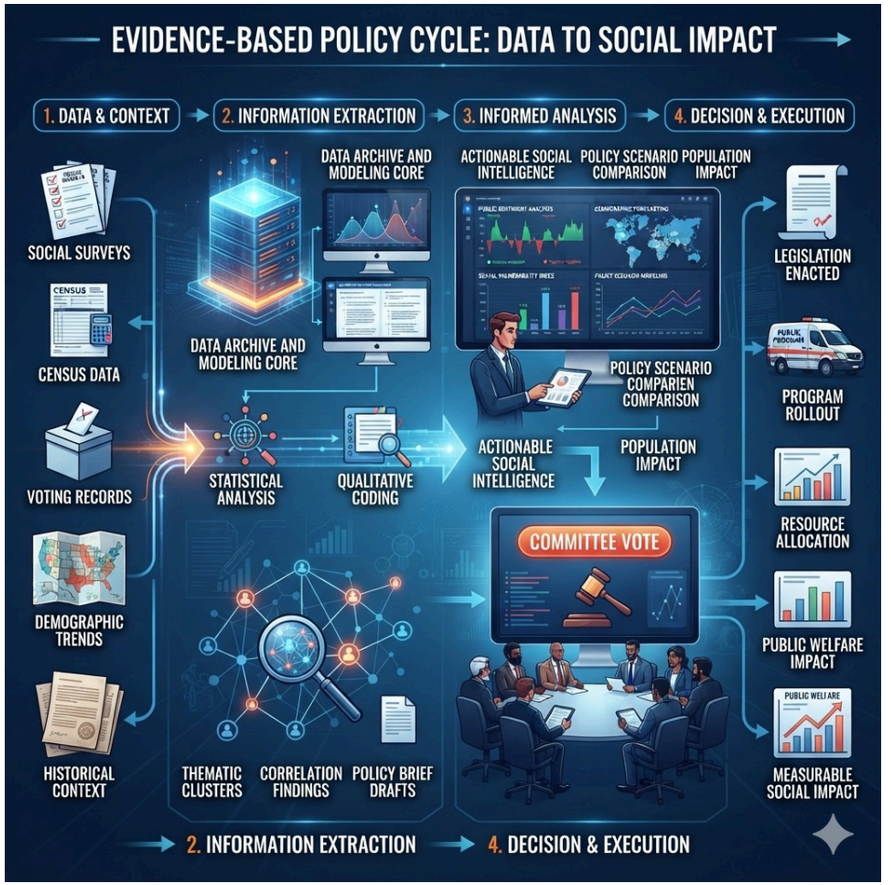
We all promise a 48-hour response time (not including weekends).

*I am far more responsive to these questions in person, in a scheduled appointment during office hours, than over email. Office hours are Fridays from 14:00 to 15:00 virtually.

We take data from the world…

and we produce:

- **Descriptive information** - summary statistics, plots, maps
- **Predictions** about the future or new contexts
  - Who will win the next election? Will the economy grow next year?
- **Evidence for causal theories** in social science
  - Are democracies less likely to go to war?
- **Policy/ Decision recommendations** - taxes, regulations, interventions



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Three (+2) Sets of Skills

1. **Computational** - how to deal with your data
    - Data structures, algorithms, software engineering
2. **Mathematical** - how to analyze your data
    - Machine learning, statistics, causal inference
3. **Substantive** - what to make of your analysis
    - Policy, law, economics, domain knowledge

AI - how to use it

- responsibly / ethically / safely / compliantly
- efficiently

Storytelling - get your message across

- data, assumptions, models and limitations

But the Library Already Does It…

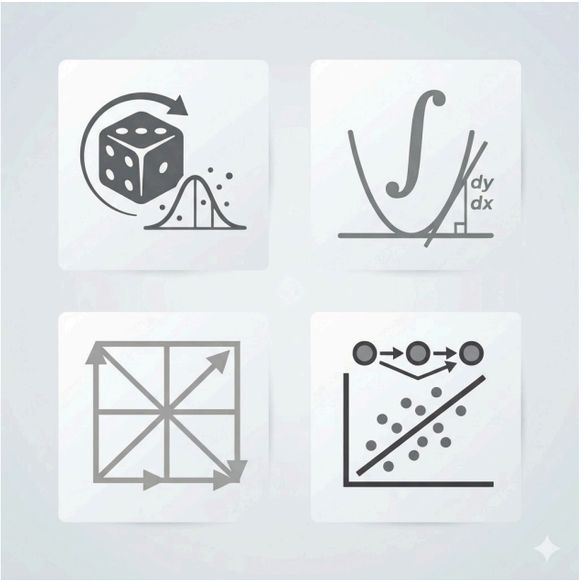
`model.fit(X, y)` is one line. So why spend a semester on the math?

- **When it breaks, only the math tells you why** - singular matrix, unidentifiable parameter, exploding gradient, a loss that will not go down
- **Every method ships with assumptions** - independence, finite variance, linearity. They are mathematical statements, and choosing a tool means checking them
- **A prediction without an error bar is an opinion** - quantifying uncertainty *is* probability theory
- **Nonsense is not flagged by the API** - leakage, spurious correlation, Simpson's paradox, a p-value mined from 200 tries
- **Libraries have a half-life of ~5 years, theorems do not** - the next framework is still a gradient and a matrix underneath
- **You have to defend the result** - to a manager, a regulator, a referee

Purpose of This Course

The mathematical foundations of data science:

Topic	Why it matters
Probability theory	Theory of the data-generating process
Calculus	How things change; optimization
Linear algebra	How to manipulate data as vectors/matrices
Statistical learning	Fitting models to data



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Probability Theory

Theory about the **process by which our data is generated**.

- Every dataset has **signal** (systematic patterns) and **noise** (stochastic variation)
- Two compatible views of randomness:
  - *Truly random* - quantum fluctuations, “coin flips”
  - *Deterministic but unobserved* - we don’t model every cause
- End goal: **extract the signal, quantify the noise**

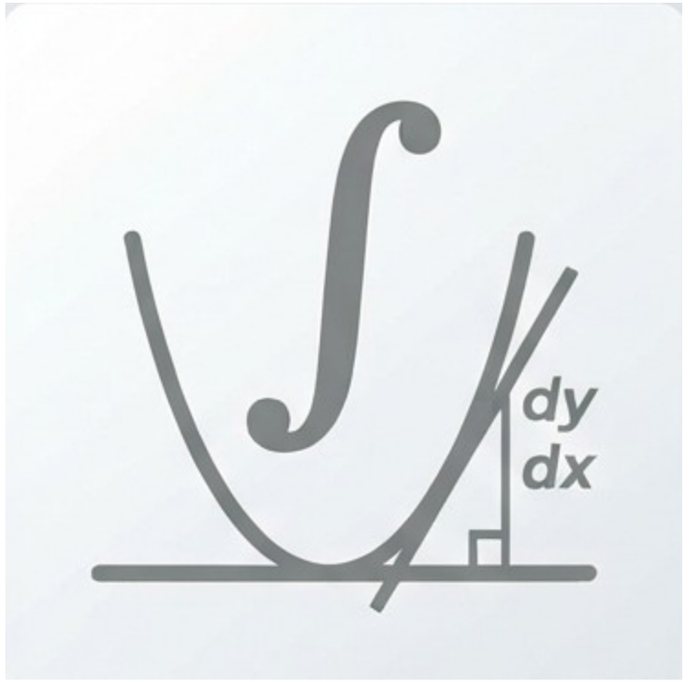


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Calculus

Mathematics of functions and how things **change**.

- *How does x change with y?* - description, prediction, theory
- *What model best fits the data?* - optimization is calculus
- Needed for probability theory with **continuous random variables**
  - Functions map events to probabilities via integrals



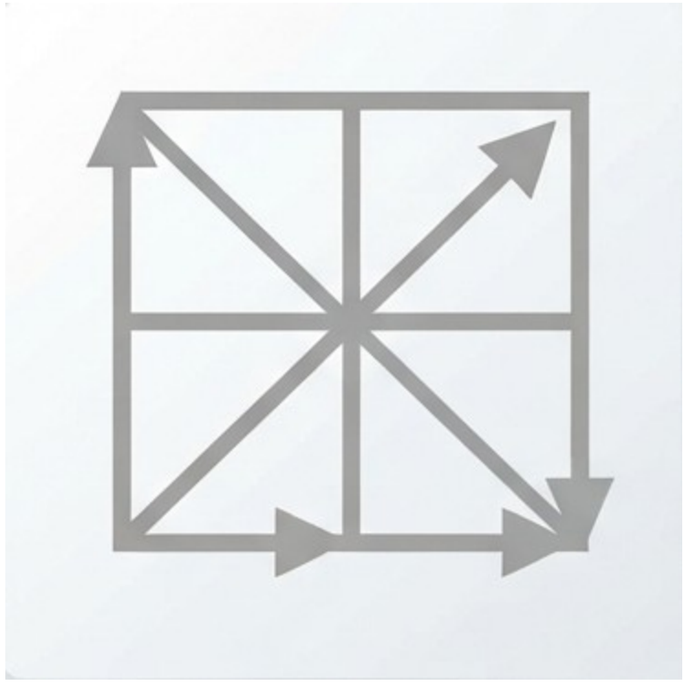
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Linear Algebra

How to manipulate **vectors and matrices** - which is how you manipulate data.

$$\begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix}$$

Each row is an observation. Each column is a feature.

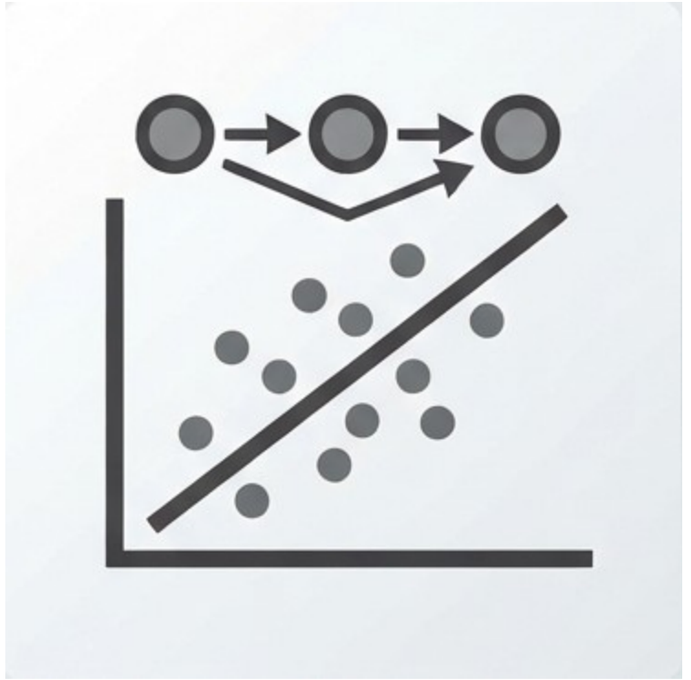


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Statistical Learning

Putting it all together:

- Combine theory about the data-generating process (**probability**)
- with tools to manipulate functions (**calculus**) and datasets (**linear algebra**),
- to **fit models to data**
- in order to **describe**, **predict**, and **theorize about** the world



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Part I: Sample Spaces & Counting

The garden of forking data

Let S be the **sample space** - the set of all possible **outcomes** of an experiment.



single die roll:

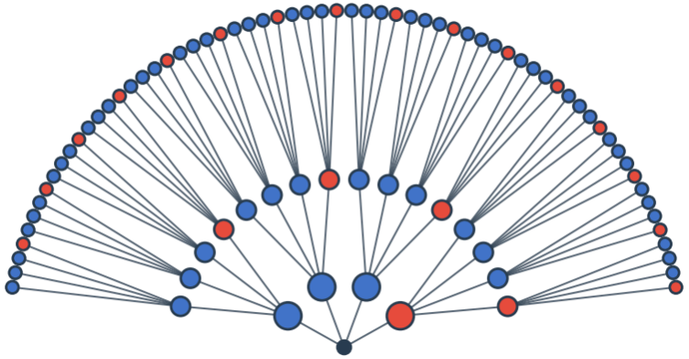
sample space 6 **outcomes** (1, 2, 3, 4, 5, 6)

The 4 sided dice with two colors

A four sided dice with 3 blue sides and one red.

4 sided dice - three tosses

sample space: 64 **outcomes**





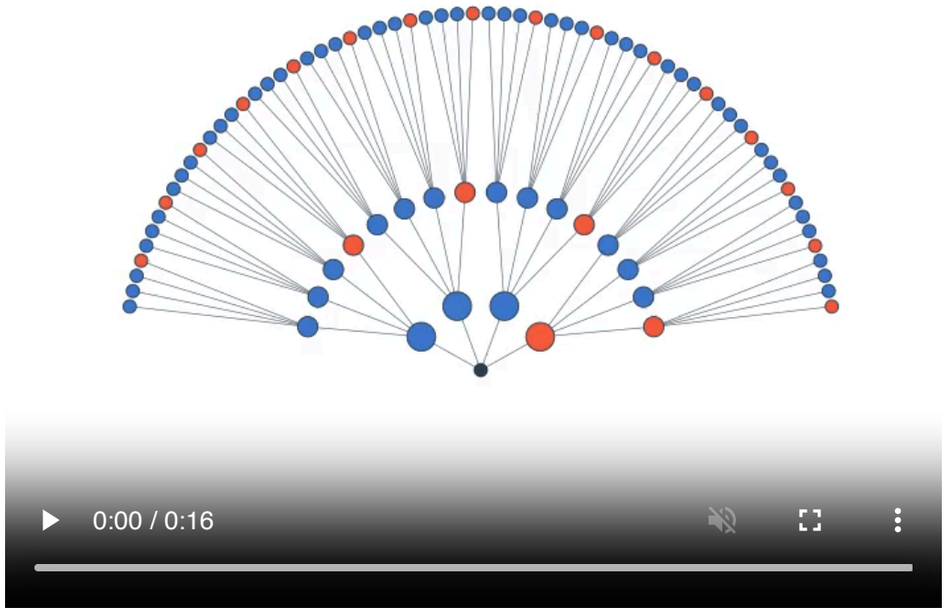
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Multiplication Rule

If sub-experiment A has a outcomes and sub-experiment B has b outcomes independently, the compound experiment has ab outcomes.

From Garden to Peeble World

Think of each outcome as a peeble so that we can imagine the sample space as a collection of peebles.



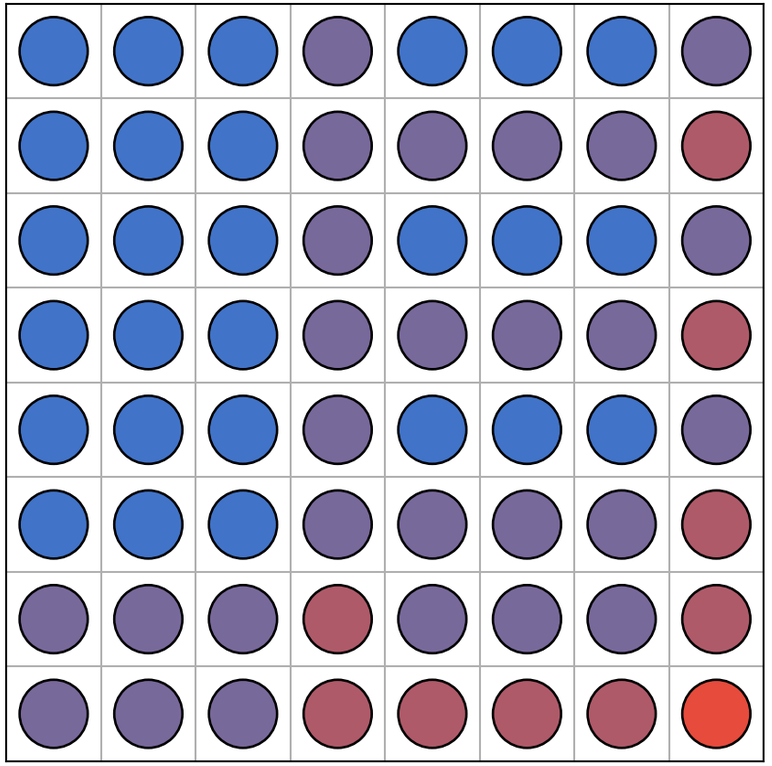
Events

Let S be the **sample space** - the set of all possible outcomes of an experiment.

Example: Three tosses of 4-sided die

4 events - order does not matter for the definition of the event

- `3 Blue`
- `2 Blue & 1 Red`
- `1 Blue & 2 Red`
- `3 Red`



Events

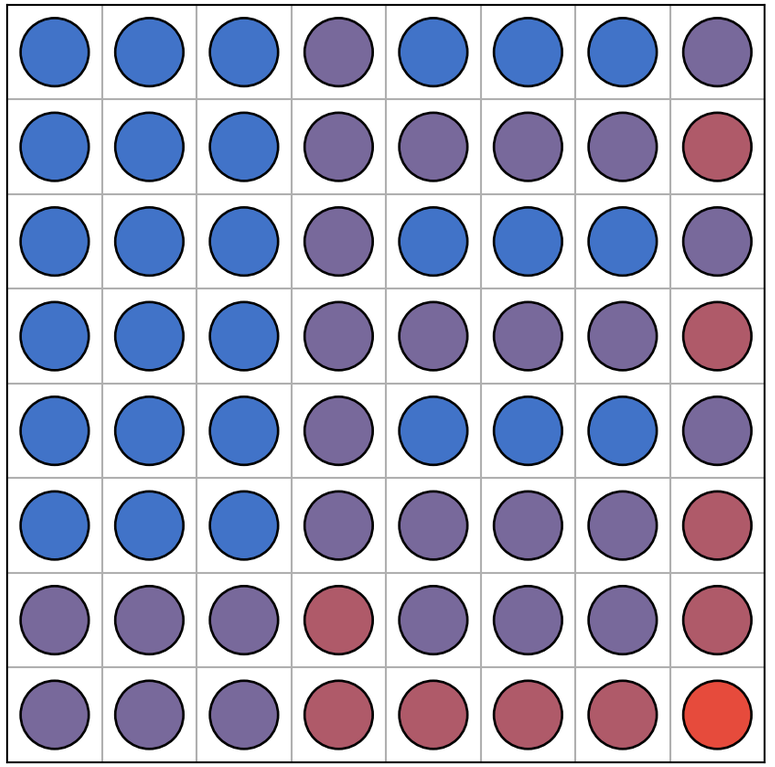
An event $A \subseteq S$ is any subset of the sample space.

Operation	Symbol	Meaning
Union	$A \cup B$	A **OR** B occurs
Intersection	$A \cap B$	*Both* A **AND** B
Complement	A^c	A does **NOT** occur

Boolean algebra

These are the same logical operators such as conjunction (and), disjunction (or), and negation (not) known from boolean algebra.

Try it on pebble world



De Morgan's Laws

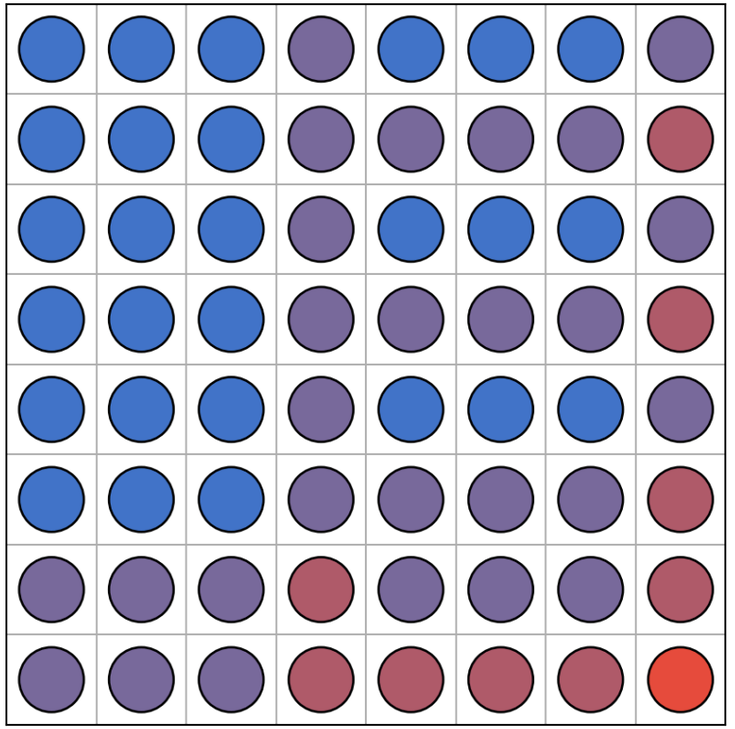
$$(A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

In plain language:

- **NOT** (A **OR** B) = **NOT** A **AND NOT** B
- **NOT** (A **AND** B) = **NOT** A **OR NOT** B

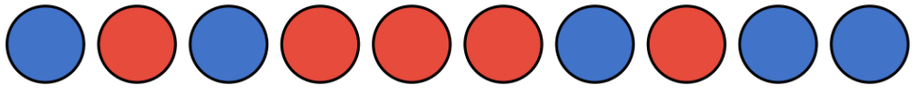
Try it on pebble world

Verify both laws: identify which pebbles satisfy each side. Do the sets match?



Building Notation: 10 Tosses

(Example 1.2.2) Our two colored dice is tossed 10 times.



$$S = \{(s_1, s_2, \dots, s_{10}) : s_j \in \{B, R\}\}$$

Let A_n = the n-th flip is Blue. Then:

$$B = \bigcup_{n=1}^{10} A_n \quad \text{(at least one Blue)}$$

$$C = \bigcap_{n=1}^{10} A_n \quad \text{(all Blue)}$$

$$D = \bigcup_{n=1}^{9} (A_n \cap A_{n+1}) \quad \text{(at least two consecutive Blue)}$$

Naive Definition of Probability

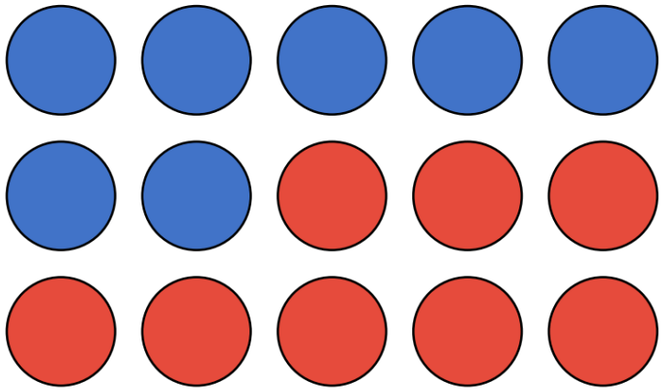
$$P_{\text{naive}}(A) = \frac{|A|}{|S|} = \frac{\text{outcomes favorable to } A}{\text{total outcomes}}$$

Immediately:

$$P_{\text{naive}}(A^c) = 1 - P_{\text{naive}}(A)$$

> ⚠ **What's "naive" about it?**
>
> The assumption of **equal probability on every outcome** must be justified by the experiment. Tossing a fair coin – yes. The color of our two colored dice – only if we keep track of the sides not just the colors. Selecting a random voter – maybe not.

$$P_{\text{naive}}(A) = \frac{7}{15}$$



Leibniz's Mistake

Two fair dice (🎲) - is a sum of **11** or a sum of **12** more likely?

Ways to get sum 11: $(5, 6)$ and $(6, 5)$ - **2 outcomes** (dice are distinguishable)

Ways to get sum 12: $(6, 6)$ - **1 outcome**

By the multiplication rule: $6 \times 6 = 36$ total equally likely outcomes.

$$P(\text{sum} = 11) = \frac{2}{36} = \frac{1}{18}, \qquad P(\text{sum} = 12) = \frac{1}{36}$$

Always label your objects

Leibniz famously claimed both sums were equally likely - because he forgot the dice are distinguishable.

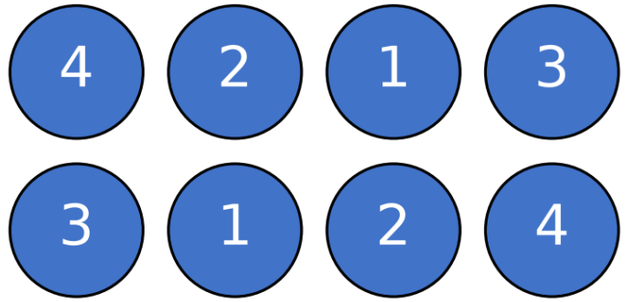
Permutations and Factorials

A **permutation** of $\{1, 2, \dots, n\}$ is an arrangement in some order.

How many permutations of n objects? Set $k = n$ above:

$$n \cdot (n-1) \cdots 1 = n!$$

For instance: 3, 1, 2, 4 is one permutation of $\{1, 2, 3, 4\}$; there are $4! = 24$ in total.



The Birthday Problem

With k people in a room, what is $P(\text{at least two share a birthday})$?

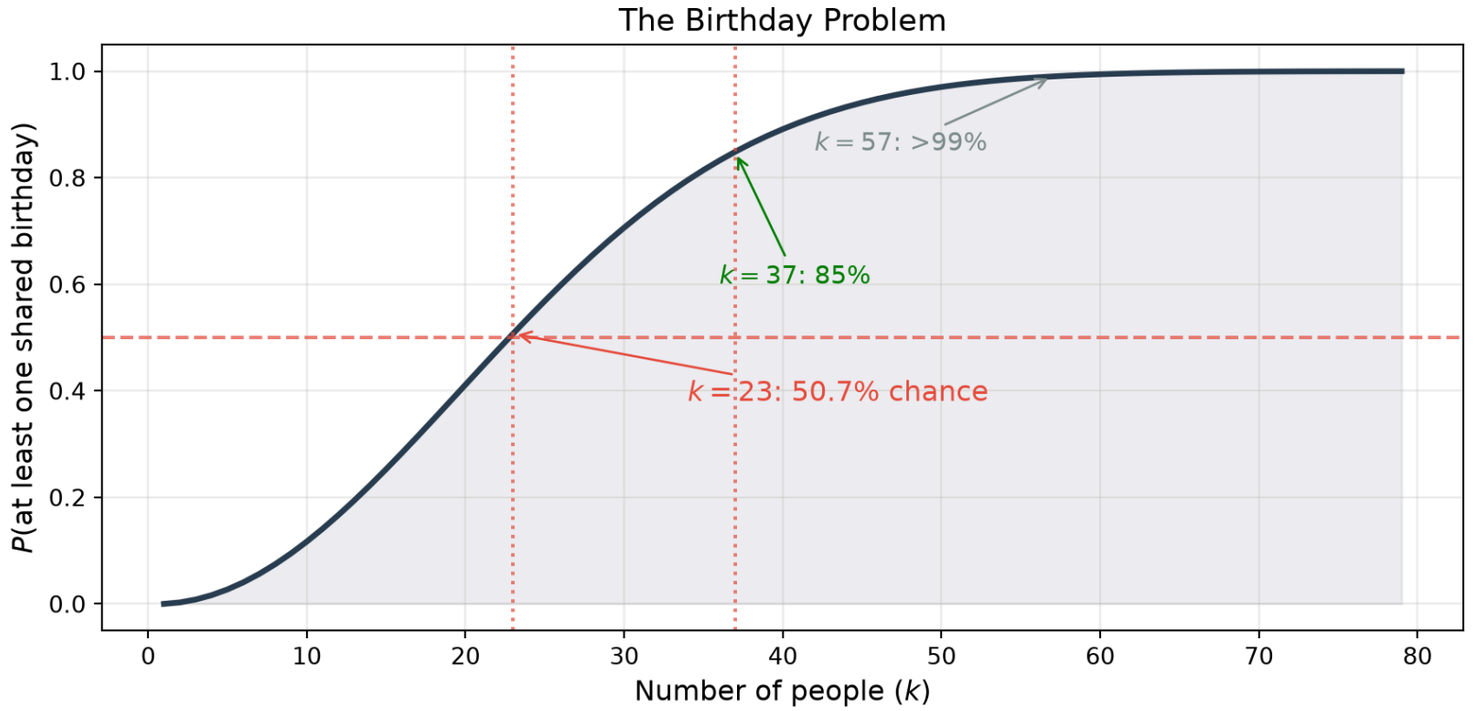
> **Count the complement**
>
> It is easier to compute $P(\text{no match})$ and subtract.

No shared birthdays = sampling 365 days **without replacement** across k people:

$$P(\text{no match}) = \frac{365 \cdot 364 \cdots (365 - k + 1)}{365^k}$$

$$P(\text{match}) = 1 - \frac{365 \cdot 364 \cdots (365 - k + 1)}{365^k}$$

Birthday Problem: The Curve



Combinations (Binomial Coefficient)

When **order doesn't matter**, use:

$$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!} = \frac{n!}{(n-k)!\,k!}$$

"n choose k" - the number of size-k **subsets** of a size-n set.

Example: 2-person committee from 4 people:

$$\binom{4}{2} = \frac{4 \cdot 3}{2!} = 6$$

> **Tip**
>
> The left form $\frac{n(n-1)\cdots(n-k+1)}{k!}$ is easier for large n: $\binom{100}{2} = \frac{100 \cdot 99}{2} = 4{,}950$

Counting in the 4 sided die example

The labels we have are:

- $N = \{2, 3, 4\}$ identifier of the blue ball (i.e. the number on the die, the red side carries the number 1)
- $T = \{\text{left}, \text{right}, \text{top}\}$ identifier of postion or corner of triangle (for our 3 toss example each corner of the triagular arrangement corresponds to the first, second and third toss).

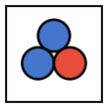


Let consider one specific sequence: blue, blue, red and we fill the corner of the triagle clockwise starting at the left corner.

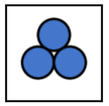
We use the multiplication rule: to get the count $c_{\text{bbr}} = 3 \times 3 \times 1 = 3^2 \times 1^1 = 9$.

> **Note**
>
> You can think of this as sampling with replacement, where you sample from two urns. The first urn has 3 blue balls. The second urn has just one ball.
>
> You select one ball from the blue urn and put it back. Then you select another one (and put it back). Then you pick a ball from the red urn (and put it back).

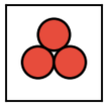
Lets look at some other examples



Different order of tosses also gives blue $c_{\mathrm{brb}} = 3 \times 1 \times 3 = 3^2 \times 1^1 = 9 = c_{\mathrm{bbr}}$



Three blue $c_{\mathrm{bbr}} = 3 \times 3 \times 3 = 3^3 \times 1^0 = 27$



Three red $c_{\mathrm{rrr}} = 1 \times 1 \times 1 = 3^0 \times 1^3 = 1$

The general solution

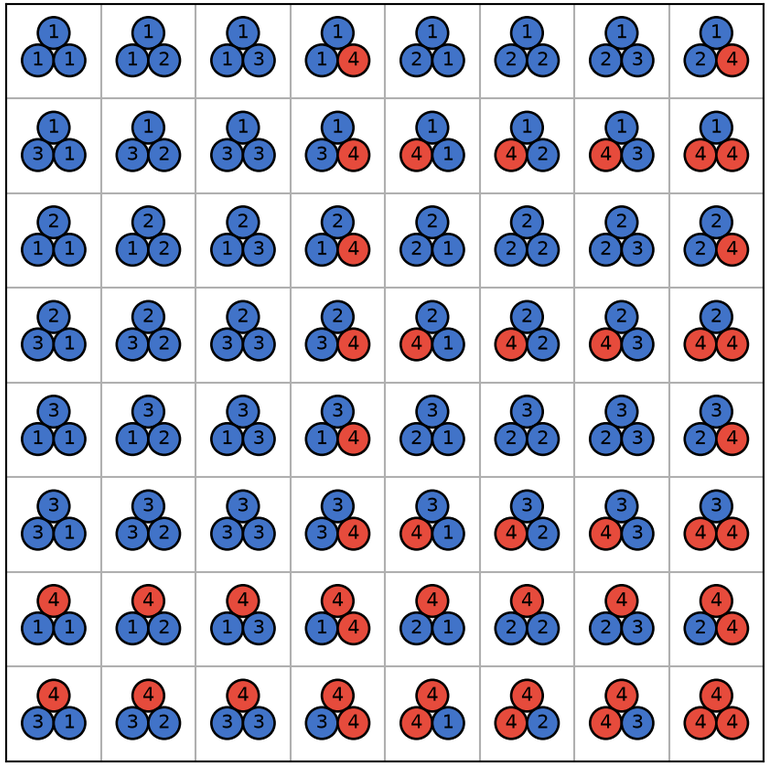
Let's call

- $t = 3$ the number of tosses
- $d = 4$ the numer of sides of the die
- $s = 3$ the numer of sides with the color blue
- b the number of blue sides per outcome (aka peeble / grid cell on the left)

then the general solution to b blue sides in t tosses for a d-sided get is

$$c(b) = s^b \times (d - s)^{(t-b)} = 3^b \times 1^{(3-b)}$$

Convince yourself:



Combinations of peebles with the same number of blue

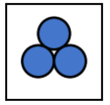
Now we consider how many ways are there to arrange the triangle such that there are two blue corners.

We have b blue that we want to position in the t corners (tosses) of the triangle but we do not care about which corner gets to be blue.

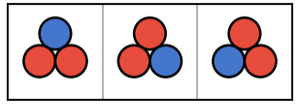
That is why there are $\binom{t}{b}$ ways to do so.

> **Note**
>
> This is a sampling without replacement where the order does not matter. Similar to the comitee example from before.

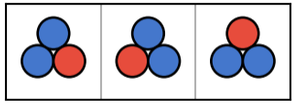
3 Blue $\Rightarrow \binom{3}{3} = 1$



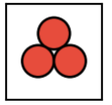
1 Blue & 2 Red $\Rightarrow \binom{3}{1} = 3$



2 Blue & 1 Red $\Rightarrow \binom{3}{2} = 3$



3 Red $\Rightarrow \binom{3}{0} = 1$

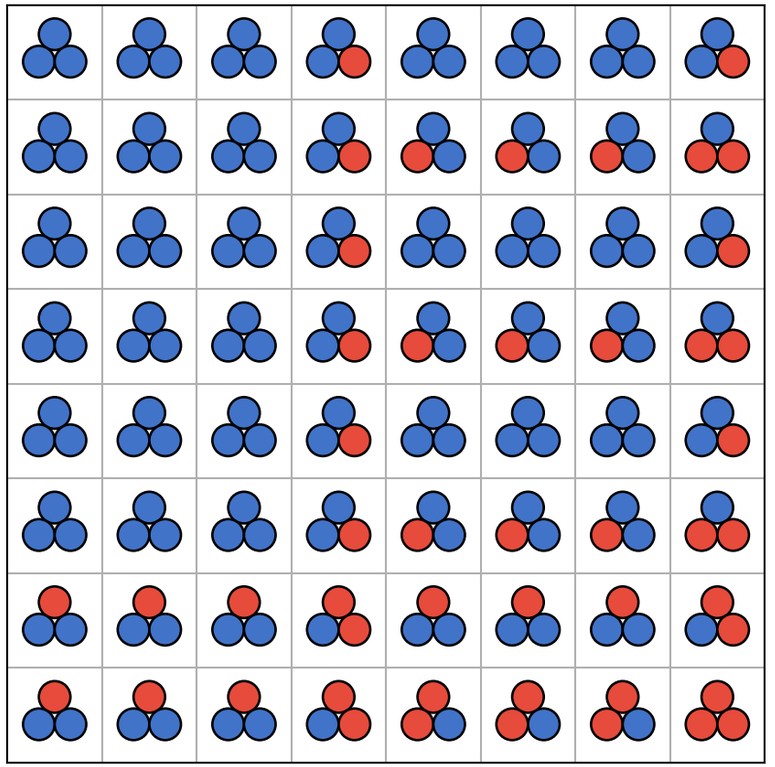


Counting events and outcomes

Say we consider events which make statements about the order at which red and blue occur. In that case the order of colors does matter for the definition of an event.

Blue	Red	Events	Outcomes per event	Outcomes
3	0	$\binom{3}{3} = 1$	$3^3 \times 1^0 = 27$	27
2	1	$\binom{3}{2} = 3$	$3^2 \times 1^1 = 9$	27
1	2	$\binom{3}{1} = 3$	$3^1 \times 1^2 = 3$	9
0	3	$\binom{3}{0} = 1$	$3^0 \times 1^3 = 1$	1
		8		**64**

Count for yourself



Counting events and outcomes

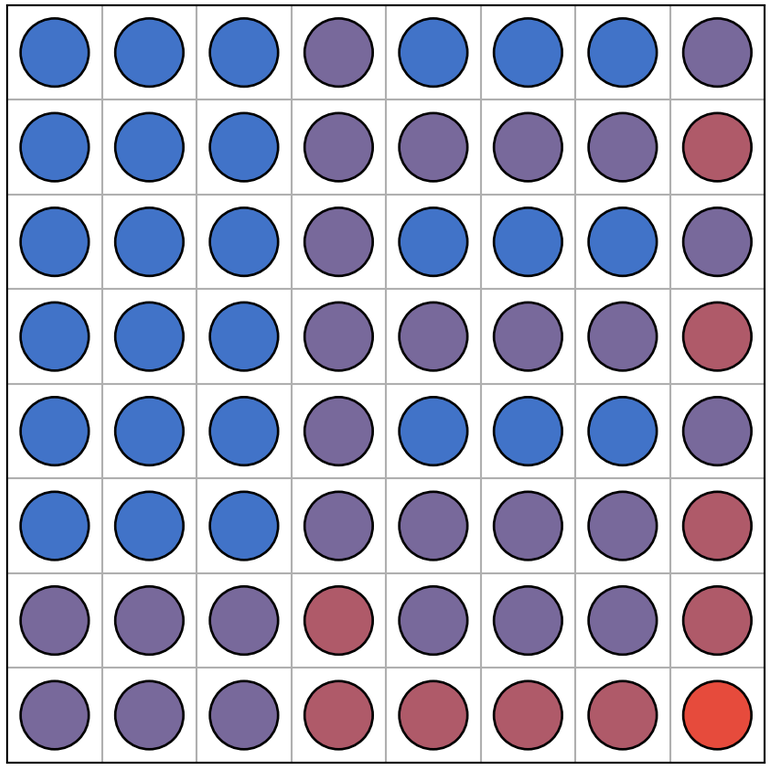
Say we consider events which do not make statements about the order at which red and blue occur. In that case the order of colors does not matter for the definition of an event.

Blue	Red	Events	Outcomes per event	Outcomes
3	0	1	$\binom{3}{3} \times 3^3 \times 1^0 = 1 \times 27$	27
2	1	1	$\binom{3}{2} \times 3^2 \times 1^1 = 3 \times 9$	27
1	2	1	$\binom{3}{1} \times 3^1 \times 1^2 = 3 \times 3$	9
0	3	1	$\binom{3}{0} \times 3^0 \times 1^3 = 1 \times 1$	1
		4		**64**

Multiplication Rule

Again we apply the multiplication rule to count the outcomes (peebles) for a given event. First, we count how many ways we can arange the blue on the triangle. Then we count how many ways are there to populate that arangement.

Count for yourself



Combinations vs Permutations

	Order matters?	Formula	Example
Sampling with replacement	Yes	n^k	How many k-digit pin codes are there with n symbols
Sampling without replacement	Yes	$\frac{n!}{(n-k)!}$	How many ways can you award k medals to n participants?
Permutation	Yes	$n!$	How many ways can I seat n people in a row
Combination without replacement	No	$\binom{n}{k} = \frac{n!}{(n-k)!\,k!}$	How many ways can you choose a k-person committee from n people?

In general, when position matters, we call it a permutation. When order does not matter, it's a combination.

Axiomatic (Non-Naive) Probability

We are now ready to define a version of probability that does not assume equal probability on all outcomes, but should still look intuitively familiar.

We define a probability space consisting of a sample space S and a probability function P which takes an event $A \subseteq S$ as input and returns $P(A)$, a real number between 0 and 1, as output. The function P must satisfy the following two axioms:

1. $P(\emptyset) = 0$ and $P(S) = 1$
2. If events $A_1, A_2, \ldots$ are **disjoint** (mutually exclusive) events, then :

$$P\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} P(A_j)$$

Any function satisfying these axioms is a valid probability function - **no equal-weight assumption required**.

Interpretation of Probability

These minimal axioms don't say what P *means*.

Frequentist view: Long-run frequency over infinite repetitions.

- Toss a fair coin infinitely → long-run frequency of Heads is $\frac{1}{2}$

Bayesian view: Degree of belief about an event, even a one-off.

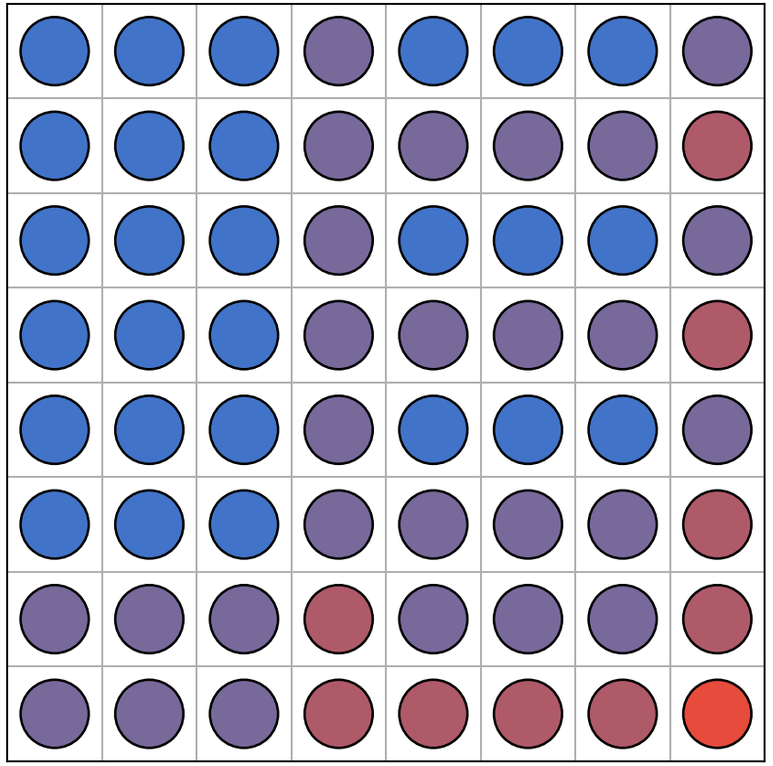
- Probability that a specific candidate wins the next election

> ⓘ **Both interpretations obey the same axioms.**
>
> The debate is about meaning, not mathematics.

Properties of Probability

From the two axioms alone:

1. $P(A^c) = 1 - P(A)$
2. If $A \subseteq B$ then $P(A) \leq P(B)$
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

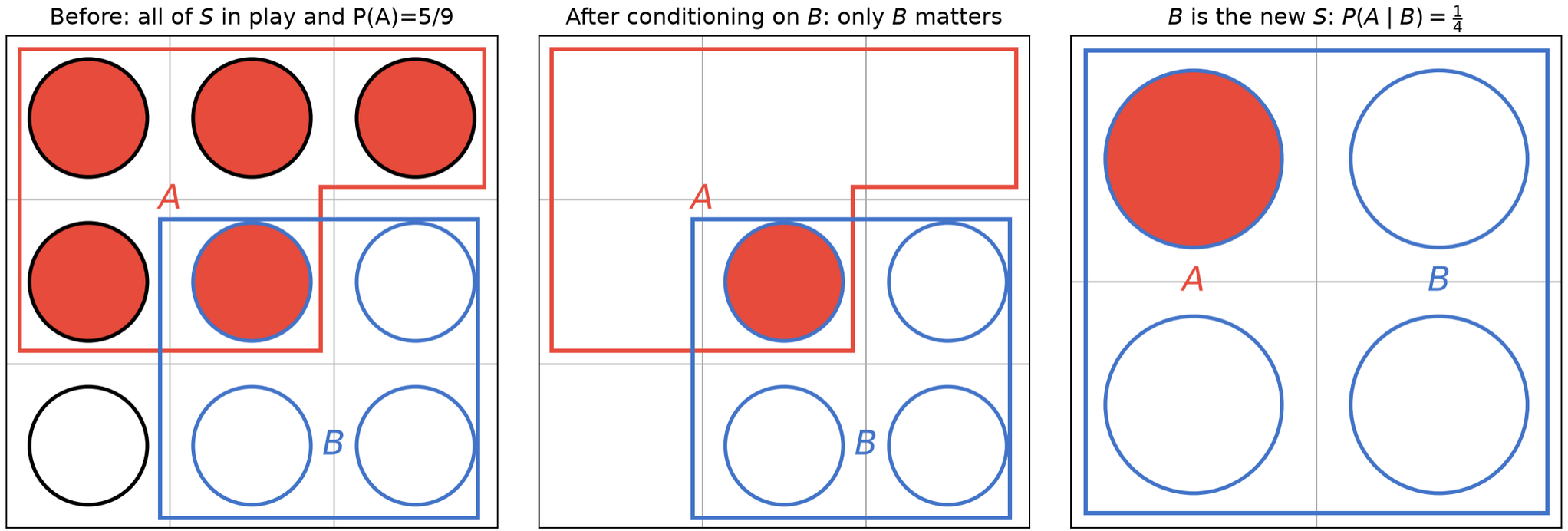


Part II: Conditional Probability

> *Conditioning is the soul of statistics.*

Conditioning Restricts the Sample Space

Definition: The probability of an event given that another event has already happened.



Learning that event B occurred **restricts the sample space** to B - we renormalize so probabilities still sum to 1 over the new space.

> ⓘ **In a Bayesian sense, *all* probabilities are conditional - they are always stated relative to some background information or model.**

Conditional Probability: Definition

Formally, if A and B are events with $P(B) > 0$, then the **conditional probability** of A given B is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

- A is the **event** we want to update
- B is the evidence we **condition on**
- $P(A)$ is the **prior** - probability before observing B
- $P(A \mid B)$ is the **posterior** - updated probability after B

Example: Drawing Cards

Example 2.2.2 (p.46):

Two cards are drawn randomly from a standard deck, one at a time without replacement.

- Let A be the event that the first card is a heart.
- Let B be the event that the second card is red.

What is $P(B \mid A)$? We can just reason about this one.

- Note that there are 52 cards in a deck, 13 of each suit (hearts, diamonds, clubs, and spades), and hearts and diamonds are red.
- After a heart is drawn, $13 + 12 = 25$ red cards remain, $52 - 1 = 51$ cards remain in the deck
- Hence, $P(B \mid A) = \frac{25}{51}$



Example: Drawing Cards

Example 2.2.2 (p.46):

Two cards are drawn randomly from a standard deck, one at a time without replacement.

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- Let B be the event that the second card is red.

What is $P(B \mid A)$? We can just reason about this one.

Verify via the formula

$P(A \cap B) = \frac{13 \cdot 25}{52 \cdot 51}$

$P(A) = \frac{1}{4}$ (13 hearts in a deck of 52)

$\Rightarrow P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{13 \cdot 25/(52 \cdot 51)}{1/4} = \frac{25}{51}$ ✓



Example: Drawing Cards

Example 2.2.2 (p.46):

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- Let A be the event that the first card is a heart.
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What about $P(A \mid B)$? That is, the event that the first card is a heart given the second card is red?



Example: Drawing Cards

Example 2.2.2 (p.46):

Two cards are drawn randomly from a standard deck, one at a time without replacement.

- Let A be the event that the first card is a heart.
- Let B be the event that the second card is red.

What about $P(A \mid B)$? That is, the event that the first card is a heart given the second card is red?

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{25/204}{1/2} = \frac{25}{102}$$



> **Note**
>
> - $P(A \cap B) = P(B \cap A)$, so $25/204$ as from before
> - $P(B) = 1/2$, this is a little tricky. But think of it from the vantage point of before having run the experiment.

Conditional probability is not symmetric in general.

⚠ $P(A \mid B) \neq P(B \mid A)$

From the cards example:

$$P(B \mid A) = \frac{25}{51} \approx 0.49 \qquad \text{vs.} \qquad P(A \mid B) = \frac{25}{102} \approx 0.25$$

Note

Even though A occurs *before* B, $P(A \mid B)$ is well-defined. Conditional probability is about **information**, not time or causation.

For further intuition: study **Examples 2.2.5, 2.2.6, 2.2.7** in Blitzstein & Hwang.

Law of Total Probability (LOTP)

Let $A_1, \ldots, A_n$ be a partition of the sample space S, meaning that the A_i's are disjoint events and their union is S, with $P(A_i) > 0$ for all i.

Then:

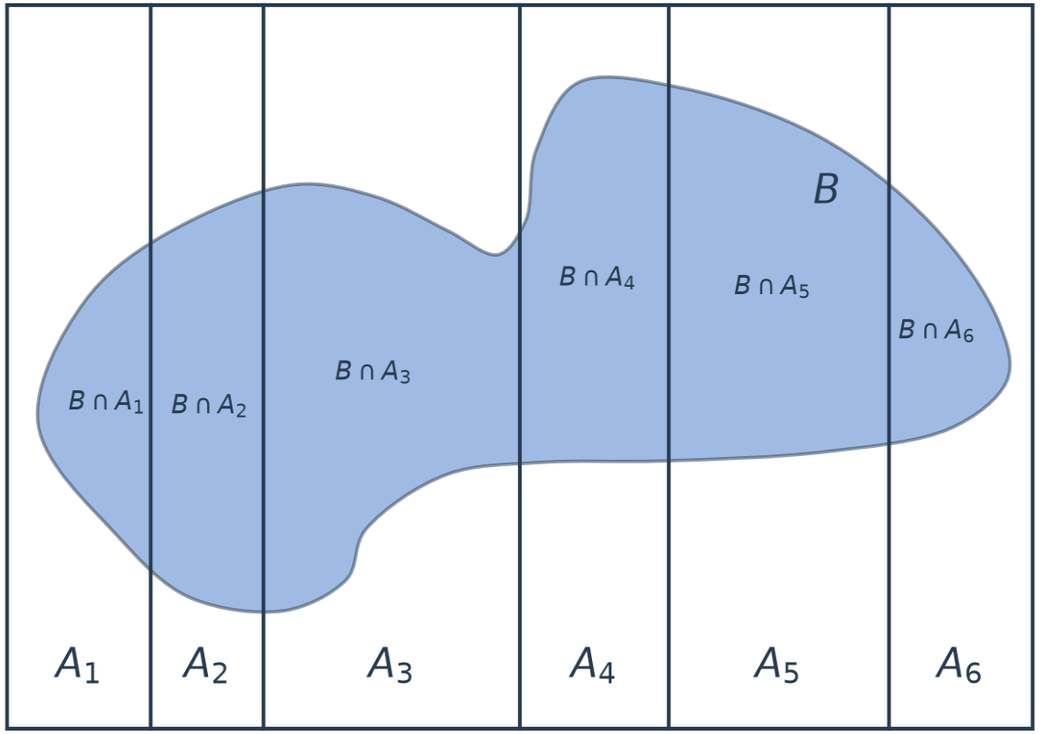
$$P(B) = \sum_{i=1}^{n} P(B \mid A_i)\, P(A_i)$$

$$= P(B \mid A_1)\, P(A_1) + \cdots + P(B \mid A_n)\, P(A_n)$$

For more intuition, work through:

- Proof of Theorem 2.3.6
- Examples 2.3.7, 2.3.8, 2.3.9, 2.3.10

This is easiest to see with a picture; see complete proof under **Theorem 2.3.6.**



After Fig. 2.3 in Blitzstein & Hwang

Practice Problems and explain videos

Counting (Ch. 1)

- Strategic Practice: Questions 1 and 2
- Exercises: 8, 24, 25, 28, 34

Conditioning (Ch. 2)

- Exercises: 1, 2, 22, 23, 25

Solutions at projects.iq.harvard.edu/stat110

Brief videos from projects.iq.harvard.edu/stat110

- Story Proofs Explained: Binomial Coefficients and Vandermonde's Identity
- Conditional Probability Explained: Visual Intuition